

$$= 0.56 \text{ m s}^{-2}$$

4 Ans: C

Angle of initial velocity = $\tan^{-1}(60 / 170) = 19.44^\circ$

Time taken to hit pigeon is when bullet has travelled horizontal displacement of 170 m.

Time taken = $170 / u_x = 170 / (300 \cos 19.44^\circ) = 0.60 \text{ s}$